

Bipartite Turán problem on cographs

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Abstract

A *cograph* is a graph that contains no induced path P_4 on four vertices or equivalently a graph that can be constructed from vertices by *sum* and *product* operations. We study the bipartite Turán problem restricted to cographs: for fixed integers $s \leq t$, what is the maximum number of edges in an n -vertex cograph that does not contain $K_{s,t}$ as a subgraph? This problem falls within the framework of induced Turán numbers $\text{ex}(n, \{K_{s,t}, P_4\text{-ind}\})$ introduced by Loh, Tait, Timmons, and Zhou.

Our main result is a *Pumping Theorem*: for every $s \leq t$ there exists a period R and R *core* cographs such that for all sufficiently large n an extremal graph is obtained by repeatedly *pumping* one designated pumping component inside the appropriate core (depending on $n \bmod R$). Moreover, we determine the linear coefficient of $\text{ex}(n, \{K_{s,t}, P_4\text{-ind}\})$ to be $s - 1 + \frac{t-1}{2}$. Moreover, the pumping components are $(t-1)$ -regular and have $s-1$ common neighbours in the core graphs, giving a particularly rigid extremal star-like shape.

Motivated by the rarity of complete classification of extremal configurations, we completely classify all $K_{3,3}$ -free extremal cographs by proof. We also develop a dynamic programming algorithm for enumerating extremal cographs for small n .

1 Introduction

1.1 Background

The Turán problem is one of the central questions in extremal graph theory: given a graph H , what is the maximum number of edges $\text{ex}(n, H)$ in an n -vertex graph that does not contain H as a subgraph? Turán's theorem [23] determines this exactly when $H = K_{r+1}$ is a clique: $\text{ex}(n, K_{r+1}) = \left(1 - \frac{1}{r}\right) \frac{n^2}{2} + O(1)$, with the unique extremal graph being the *Turán graph* $T_r(n)$ —the complete r -partite graph with parts as equal as possible. Notably, $T_r(n)$ as a complete multipartite graph is a cograph, so in the case of cliques the extremal construction already lies within the class of cographs. For general graphs H , the Erdős-Stone-Simonovits theorem [23] provides the asymptotic answer: $\text{ex}(n, H) = (1 - 1/(\chi(H) - 1) + o(1)) \binom{n}{2}$. However, when H is bipartite (i.e., $\chi(H) = 2$), this theorem only yields $\text{ex}(n, H) = o(n^2)$, leaving the exact order of magnitude as a challenging open problem [11].

For complete bipartite graphs $K_{s,t}$ with $s \leq t$, the Turán problem is known as the *Zarankiewicz problem*, named after the Polish mathematician who posed it in the 1950s [16]. The fundamental upper bound was established by Kővári, Sós, and Turán [18]:

$$\text{ex}(n, K_{s,t}) \leq \frac{1}{2}(t-1)^{1/s} n^{2-1/s} + \frac{1}{2}(s-1)n. \quad (1)$$

Loh, Tait, and Timmons [19] introduced the *induced Turán number* $\text{ex}(n, \{H, F\text{-ind}\})$, defined as the maximum number of edges in an n -vertex graph containing no copy of H as a subgraph and no induced copy of F . Our problem falls within this framework: we study $\text{ex}(n, \{K_{s,t}, P_4\text{-ind}\})$, the bipartite Turán problem restricted to cographs.

1.2 Cographs

A *cograph* (short for *complement-reducible graph*) is a graph that contains no induced path on four vertices (P_4) [7]. Equivalently, cographs are precisely the graphs that can be constructed from single vertices using two recursive operations: the *product* (also called *join*) and the *sum* (disjoint union). Both operations are associative and commutative.

For two graphs G_1, G_2 , define their *product* $G_1 \times G_2$ as their disjoint union together with all edges between $V(G_1)$ and $V(G_2)$:

$$E(G_1 \times G_2) = E(G_1) \cup E(G_2) \cup \{(v_1, v_2) : v_1 \in V(G_1), v_2 \in V(G_2)\}.$$

Define their *sum* $G_1 + G_2$ as simply their disjoint union. Iterating these operations for graphs $G_1, \dots, G_n$, we define $\prod_{j \in [n]} G_j$ and $\sum_{j \in [n]} G_j$ respectively.

1.3 Main results

For $s, t \in \mathbb{N}$ with $s \leq t$, we call a cograph (s, t) -*extremal* if it is edge-maximal among cographs not containing $K_{s,t}$. To build some first intuition about the problem, we developed a dynamic programming framework (Section 2.1) to enumerate extremal cographs for small n . Check out the presentation of found extremal (s, t) -extremal cographs at <https://extremal-cographs.fly.dev>. All code is available at https://github.com/JayPiZimmermann/zarankievicz_cographs.git

Definition 1 (Pumping). Let G be a graph and let $H \subseteq G$ be a subgraph. We say that we k times *pump* (or k times *pump up*) H in G to obtain the graph G' constructed by adding k copies of H , where each copy has every vertex connected to the $V(G - H)$ -neighbourhood of its corresponding vertex in H . Formally:

$$G' := \left(V(G) \setminus V(H) \cup \bigcup_{j \in [k+1]} V(H_j), E(G - H) \cup \bigcup_{j \in [k+1]} \left(E(H_j) \cup \bigcup_{v \in V(H)} \{v'\} \times \mathcal{N}_{V(G-H)}(v) \right) \right), \quad (2)$$

where $H_1, \dots, H_{k+1}$ are copies of H and vertex $v \in V(H)$ corresponds to vertex v' in $V(H_j)$.

Our first main result shows that for large n , extremal cographs arise by pumping a specific component inside a core graph.

Theorem 1 (Pumping Theorem). Let $s, t \in \mathbb{N}$ with $2 \leq s \leq t$. There exist $N_{s,t}, R \in \mathbb{N}$ and core graphs $G_0, \dots, G_R$ each of size at most $N_{s,t}$ with designated $(t-1)$ -regular components $H_j \subseteq G_j$ with $s-1$ common neighbors outside of H_j such that for any n large enough there is an (s, t) -extremal cograph on n vertices obtained by pumping $H_{(n \bmod t)}$ inside of $G_{(n \bmod R)}$. That is, for n large enough

$$\text{ex}(n, \{K_{s,t}, P_4\text{-ind}\}) = a_{(n \bmod R)} + \left(\frac{t-1}{2} + s - 1 \right) \cdot n \quad (3)$$

for constants $a_0, \dots, a_{R-1} < 0$. Moreover, there exists at least one residue class $g \in \mathbb{Z}/R\mathbb{Z}$ such that for large n with $n \bmod R = g$ the (s, t) -extremal cograph on n vertices is connected.

The key structural tool underlying the Pumping Theorem is the *Lifting Theorem* (Theorem 5), which shows that extremal cographs admit a specific decomposition into products of small cores with sums of smaller graphs. This lifting procedure controls the connectivity components of extremal graphs and constrains their structure, enabling the eventual pumping behavior. In Theorem 7 we also prove a pumping-like behaviour of cographs edge-maximal with respect to not containing any of a set of bicliques.

In the following three theorems, we can fully classify all extremal cographs - also for small n . For forbidden stars $K_{1,t}$, the extremal cographs are regular in most cases.

Theorem 2 (Forbidden stars). Let $t, n \in \mathbb{N}$ with $n \geq t$. In case n is even or $t-1$ is even and $2t \neq n+1$, any $(1, t)$ -extremal cograph on n vertices is $(t-1)$ -regular. Otherwise, it is the sum of a $(t-1)$ -regular cograph and a connected remainder of size at most $2t-3$.

For $K_{2,t}$, the extremal cographs have a particularly simple structure:

Theorem 3. Let $t \in \mathbb{N}$ with $t \geq 2$ and let G be a $(2, t)$ -extremal cograph on $n \geq 2$ vertices. Then, G is the product of a vertex with another cograph G_2 that is $(1, t)$ -extremal.

Our final main result provides a complete classification in the fundamental case $K_{3,3}$:

Theorem 4. Let G be an edge-maximal cograph on $n \geq 2$ vertices that does not contain $K_{3,3}$. Then, G is the product of a two-vertex cograph G_1 with another cograph G_2 . Moreover, there is always an edge-maximal cograph on $n \geq 2$ vertices such that G_1 is an edge and G_2 is the sum of cliques.

The proofs of these theorems are presented in Sections 2 and 3.

1.4 Further background and related work

More insights about the Zarankievicz problem. The Kővári-Sós-Turán bound (1), known as the *KST bound*, is known to be tight in several cases. The cases $s = t = 2$ and $s = t = 3$ were resolved by classical constructions [11]. Kollár, Rónyai, and Szabó [17] introduced *norm graphs* to show tightness when $t \geq s! + 1$, which Alon, Rónyai, and

Szabó [1] improved to $t \geq (s-1)! + 1$. Conlon [6] developed a quantitative variant of the random algebraic method to prove that $\text{ex}(K_{n,m}, K_{s,t}) = \Omega(mn^{1-1/s})$ for any $m \leq n^{t^{1/(s-1)}/s(s-1)}$, where $\text{ex}(G, H)$ is the maximal number of edges a subgraph of G can have without containing H , showing tightness of the KST bound over a broader range than previously known. However, determining the exact order of magnitude for general s and t remains a central open problem [11, 16]. For small parameters, Tan [25] used SAT solvers to compute exact values of the Zarankiewicz function, correcting errors in earlier hand-computed tables and extending the known range of parameters.

Structural constraints and improved bounds. A key insight in modern extremal combinatorics is that additional structural constraints on the host graph can yield dramatically improved bounds. A foundational result in this direction is due to Bonamy, Bousquet, Pilipczuk, Rzażewski, Thomassé, and Walczak [4], who proved that $K_{\ell, \ell}$ -free graphs avoiding induced P_5 have degeneracy $O(\ell^3)$. Since cographs (graphs avoiding induced P_4) are a subclass of induced P_5 -free graphs, this immediately implies that $K_{\ell, \ell}$ -free cographs have at most $O(\ell^3 n)$ edges—a linear bound in n . More generally, Bonamy et al. showed that $K_{\ell, \ell}$ -free graphs avoiding long induced paths or cycles have degeneracy polynomial in ℓ , with specific exponents depending on the forbidden pattern. In a major breakthrough, Nguyen [21] resolved a 1985 problem of Gyárfás by proving that induced P_5 -free graphs are polynomially χ -bounded: there exists $d \geq 2$ such that every induced P_5 -free graph G satisfies $\chi(G) \leq \omega(G)^d$.

Füredi [10] and Sudakov–Tomon [24] developed powerful techniques for bipartite Turán problems using *neighborhood intersection arguments*, where vertex sets in extremal structures cannot share too many common neighbours. Füredi [10] used this to show that graphs avoiding L^k (the lowest three levels of the Boolean lattice) have $O(n^{3/2})$ edges, confirming a conjecture of Erdős. Sudakov and Tomon [24] extended these ideas using hypergraph methods: given a dense bipartite graph, they construct a t -uniform hypergraph on common neighborhoods and apply the Hypergraph Removal Lemma to embed forbidden subgraphs.

The survey by Keller and Smorodinsky [16] develops a unified approach to the Zarankiewicz problem via ε - t -nets—a generalization of classical ε -nets where one seeks a small family of t -tuples hitting all large hyperedges. Their key result shows that $K_{t,t}$ -free bipartite graphs with VC-dimension d have $O(n^{2-1/d})$ edges, recovering the Fox-Pach-Sheffer-Suk-Zahl bound [16] with a simpler proof. For geometric intersection graphs, they obtain much stronger bounds: intersection graphs of pseudo-discs have only $O(t^6 n)$ edges when $K_{t,t}$ -free (a *linear* bound), and axis-parallel rectangle intersection graphs achieve $O(tn \cdot \log n / \log \log n)$.

Induced Turán numbers and hereditary properties. Loh, Tait, and Timmons [19] showed that for any fixed H one has $\text{ex}(n, \{H, K_{s,t}\text{-ind}\}) = O(n^{2-1/s})$. Illingworth [14] asymptotically determined $\text{ex}(n, \{H, F\text{-ind}\})$ when H is non-bipartite and F is neither an independent set nor a complete bipartite graph, complementing the focus on complete bipartite forbidden induced subgraphs. Nikiforov, Tait, and Timmons [22] proved spectral strengthenings: if an induced H -free graph on n vertices has spectral radius $\lambda(G) \geq Kn^{1-1/s}$ for appropriate K , then it contains an induced copy of $K_{s,t}$.

The study of biclique-free graphs with hereditary constraints has seen significant recent progress. Hunter, Milojević, Sudakov, and Tomon [13] proved that for bipartite H with maximum degree at most k on one side, graphs avoiding induced H and containing no $K_{s,s}$ have $O(s^c n^{2-1/k})$ edges, and conjectured that the dependence should match the ordinary Turán number up to a constant factor; Axenovich and Zimmermann [3] strengthened this result in bipartite host graphs, proving that for any $d \geq 2$ and any $K_{d,d}$ -free bipartite graph H where each vertex in one part has degree either at most d or full degree, with at most $d-2$ full-degree vertices in that part, one has $\text{ex}(K_{n,n}, \{K_{t,t}, H\text{-ind}\}) = o(n^{2-1/d})$. Here, $\text{ex}(G, \{F, H\text{-ind}\})$ is the maximal number of edges of a subgraph of G that does not contain a copy of F and no induced copy of H . This result was built on insights of Janzer and Pohoata [15], who showed that $K_{k,k}$ -free bipartite graphs with VC-dimension at most d have $o(n^{2-1/d})$ edges, improving the KST bound when the VC-dimension is smaller than the biclique parameter.

VC-dimension, cographs, and the Erdős-Hajnal conjecture. The *VC-dimension* of a graph G is defined as the VC-dimension of the hypergraph formed by closed neighborhoods $\{\mathcal{N}(v) : v \in V(G)\}$. Intuitively, a graph has bounded VC-dimension when its neighborhood structure is not too complex—specifically, when no large vertex set can be “shattered” by neighborhoods. Bousquet, Lagoutte, Li, Parreau, and Thomassé [5] proved a fundamental dichotomy: a hereditary graph class has finite VC-dimension if and only if it excludes some bipartite graph, some co-bipartite graph, and some split graph. Remarkably, P_4 is simultaneously bipartite, co-bipartite, and split, so *forbidding P_4 alone as an induced subgraph suffices to ensure bounded VC-dimension*. Indeed, all graphs G such that the G -free graphs have bounded VC-dimension are induced subgraphs $G \subseteq P_4$.

Nguyen, Scott, and Seymour [20] resolved the Erdős-Hajnal conjecture for graphs of bounded VC-dimension, proving that every n -vertex graph with VC-dimension at most d contains a clique or stable set of size at least n^{c_d} for some

$c_d > 0$. It was a well-known simple fact that cographs satisfy the Erdős-Hajnal property with $c = 1/2$, meaning every n -vertex cograph contains a clique or independent set of size $\Omega(\sqrt{n})$ [8].

The connection between induced subgraph avoidance and density bounds is highlighted by several fundamental results. Fox and Sudakov [9] proved subexponential improvements to Rödl's theorem for hereditary graph classes, avoiding the tower-type bounds from the regularity lemma and conjecturing that polynomial bounds should hold. Gishboliner and Shapira [12] showed that induced P_4 -free graphs (cographs) have polynomial Rödl bounds, exploiting the fact that P_4 is simultaneously bipartite, co-bipartite, and split. Alon, Fischer, and Newman [2] proved that bipartite graph properties characterized by a finite set of forbidden induced subgraphs are efficiently testable with polynomial query complexity—specifically, one can ε -test membership in such a class using $\text{poly}(1/\varepsilon)$ queries, a dramatic improvement over the tower-type bounds that arise from the Szemerédi regularity lemma.

2 Biclique-Profiles

For a graph G let us define its *biclique-sequence* by $\mathcal{S}(G)$, where for $s \in \mathbb{N}$

$$\mathcal{S}_s(G) = \max \{t \in \mathbb{N} \mid K_{s,t} \subseteq G\}. \quad (4)$$

Here, we define $K_{0,t}$ as the empty graph on t vertices, i.e., $\mathcal{S}_0(G) = |G|$. Observe that $\mathcal{S}_1(G)$ is the maximum degree.

Moreover, we define the maximum of the empty set as $-\infty$. That is, the biclique-sequence of a graph is monotonically decreasing and constant $-\infty$ on indices greater than its vertex count. Furthermore, biclique-sequences are *subdiagonal*, a property defined by the following two implications

$$\mathcal{S}_j(G) < l \implies K_{j,l} \not\subseteq G \implies \mathcal{S}_l(G) < j, \quad (5)$$

$$\mathcal{S}_j(G) \geq l \implies K_{j,l} \subseteq G \implies \mathcal{S}_l(G) \geq j. \quad (6)$$

We might interpret $\mathcal{S}_\infty$ as the limit of the sequence, and deduce the following equivalence.

$$l = \mathcal{S}_0(G) < \infty \iff \mathcal{S}_{l+1}(G) = -\infty \quad (7)$$

Definition 2 (Biclique-profile). A *biclique-profile* is any decreasing subdiagonal sequence $\mathcal{P}$ in $\mathbb{N} \cup \{\infty, -\infty\}$. Observe that a biclique-profile has $\mathcal{P}_0 < \infty$ if and only if it is constant $-\infty$ from some index onward.

Let us say a graph G *fulfills* a biclique-profile $\mathcal{P}$ in case that pointwise $\mathcal{S}(G) \leq \mathcal{P}$. Furthermore, a graph is *extremal with respect to* $\mathcal{P}$ in case it fulfills $\mathcal{P}$ and it has the maximal edge count among all graphs with the same vertex count that fulfill $\mathcal{P}$. We use the same notion for the extremal cograph problem.

The set of all biclique-profiles together with the element-wise ordering form a lattice with minimal element zero and the maximal element infinity.

For a biclique-profile $\mathcal{P}$ let us define its *start index* by the minimal $i \in \mathbb{N}_0$ such that $\mathcal{P}_i < \infty$. Similarly, we define its *end index* as the minimal index where it attains its limit. When considering extremal cographs not containing $K_{s,t}$ for $s \leq t$ observe that the corresponding biclique-profile has start index s , where it attains the value $t-1$, and stop index t where it jumps to its limit $s-1$.

2.1 Dynamic programming

The recursive structure of cographs via sum and product operations naturally leads to a dynamic programming approach for computing extremal graphs. We first observe that biclique-sequences combine predictably under these operations.

Lemma 1 (Profile operations). Let G_1, G_2 be cographs on n_1, n_2 vertices respectively. Then:

- (i) For the sum $G_1 + G_2$:

$$\mathcal{S}_s(G_1 + G_2) = \max(\mathcal{S}_s(G_1), \mathcal{S}_s(G_2)) \quad (8)$$

- (ii) For the product $G_1 \times G_2$:

$$\mathcal{S}_s(G_1 \times G_2) = \max_{a+c=s} (\mathcal{S}_a(G_1) + \mathcal{S}_c(G_2)) \quad (9)$$

Proof of Lemma 1. (i) In a sum $G_1 + G_2$, there are no edges between $V(G_1)$ and $V(G_2)$. Thus any $K_{s,t} \subseteq G_1 + G_2$ must lie entirely in G_1 or entirely in G_2 , giving the pointwise maximum.
(ii) In a product $G_1 \times G_2$, every vertex in G_1 is adjacent to every vertex in G_2 . A copy of $K_{s,t}$ can use a vertices from one side of the biclique in G_1 and $c = s - a$ vertices from the other side, with the remaining vertices coming from G_2 . The maximum over all such decompositions yields the max-convolution formula. $\square$

The key observation is that the number of edges also combines predictably:

$$\|G_1 + G_2\| = \|G_1\| + \|G_2\| \quad (10)$$

$$\|G_1 \times G_2\| = \|G_1\| + \|G_2\| + n_1 \cdot n_2 \quad (11)$$

Definition 3 (Pareto frontier). For a set of pairs $\{(\mathcal{P}_j, e_j)\}_{j \in J}$ consisting of biclique-profiles and edge counts, the *Pareto frontier* is the set of pairs $(\mathcal{P}_j, e_j)$ such that there is no other pair $(\mathcal{P}_k, e_k)$ with $\mathcal{P}_k \leq \mathcal{P}_j$ (coordinatewise) and $e_k \geq e_j$, with at least one strict inequality.

For computing extremal cographs, we observe that dominated profile-edge pairs can never contribute to an extremal graph: if $(\mathcal{P}, e)$ is dominated by $(\mathcal{P}', e')$ with $\mathcal{P}' \leq \mathcal{P}$ and $e' \geq e$, then any graph fulfilling $\mathcal{P}$ also fulfills $\mathcal{P}'$, and the dominating graph has at least as many edges.

The algorithm proceeds by dynamic programming over the vertex count n . For each n , we maintain a registry $\mathcal{R}_n$ of pairs $(\mathcal{P}, \mathcal{G}_{\mathcal{P}})$, where $\mathcal{G}_{\mathcal{P}}$ is the set of cographs on n vertices having profile $\mathcal{P}$ with maximum edge count among graphs with that profile.

Algorithm: Dynamic Programming for Extremal Cographs

1. **Base case.** Set $\mathcal{R}_1 \leftarrow \{((1, 0, \dots), \{K_1\})\}$.
2. **Inductive step.** For each $n \geq 2$:
 - (a) Initialize candidate set $\mathcal{C}_n \leftarrow \emptyset$.
 - (b) For each partition $n = n_1 + n_2$ with $1 \leq n_1 \leq n_2$:
 - (i) For each $(\mathcal{P}_1, \mathcal{G}_1) \in \mathcal{R}_{n_1}$ and $(\mathcal{P}_2, \mathcal{G}_2) \in \mathcal{R}_{n_2}$:
 - Compute $\mathcal{P}_+ = \mathcal{S}(\mathcal{P}_1 + \mathcal{P}_2)$ via (8).
 - Compute $\mathcal{P}_\times = \mathcal{S}(\mathcal{P}_1 \times \mathcal{P}_2)$ via (9).
 - Compute edge counts $e_+, e_\times$ via (10)–(11).
 - Add $(\mathcal{P}_+, e_+, G_1 + G_2)$ and $(\mathcal{P}_\times, e_\times, G_1 \times G_2)$ to $\mathcal{C}_n$ for each $G_1 \in \mathcal{G}_1, G_2 \in \mathcal{G}_2$.
 - (c) **Pareto filtering.** Reduce $\mathcal{C}_n$ to the Pareto frontier with respect to the partial order $(\mathcal{P}, e) \preceq (\mathcal{P}', e')$ iff $\mathcal{P} \geq \mathcal{P}'$ and $e \leq e'$.
 - (d) Group by profile: $\mathcal{R}_n \leftarrow \{(\mathcal{P}, \{G : (\mathcal{P}, e, G) \in \mathcal{C}_n, e = e_{\max}(\mathcal{P})\})\}$.
3. **Query.** Given a biclique-profile $\mathcal{P}$ with $\mathcal{P}_0 = n$, return graphs from $\mathcal{R}_n$ whose profiles are $\leq \mathcal{P}$ (coordinatewise) with maximum edge count.

The lattice-based optimization (step 2c) prunes dominated profile pairs at the profile level *before* expanding to individual graph pairs, reducing the combinatorial explosion. For a partition (n_1, n_2) with $|\mathcal{R}_{n_1}| = p_1$ and $|\mathcal{R}_{n_2}| = p_2$ profiles, the naive approach considers $O(p_1 \cdot p_2 \cdot g_1 \cdot g_2)$ graph pairs, where g_1, g_2 are the average number of graphs per profile. The lattice filtering first computes the antichain of non-dominated profile pairs (at most $O(p_1 \cdot p_2)$), then only expands graph pairs for surviving profile combinations.

For practical computation with a pruning constraint $K_{s,t}$, we can truncate profiles to length $s+1$ since only the first s entries determine whether a graph fulfills the constraint $\mathcal{P}_s < t$. Moreover, when only interested in (a, b) -extremal graphs we can prune all profiles $\mathcal{P}$ where $\mathcal{P}_a \geq b$.

It is easy to see that our approach is correct: In an (s, t) extremal graph all components are extremal with respect to a biclique profile that depends on the remaining graph. Thus, by combining all graphs extremal to suitable profiles, the search considers all relevant candidates for extremal cographs.

By step (c) we lose the ability to enumerate all extremal graphs. We remark that one could simply skip step (c) - at the cost of significantly more memory consumption and run time. However, it is easy to see that also with step (c) one still finds at least one extremal graph for each $n \in \mathbb{N}$.

2.2 The Lifting Theorem

We say a biclique-profile $\mathcal{P}$ *dominates* another profile $\mathcal{P}'$ if $\mathcal{P}_i \geq \mathcal{P}'_i$ for all $i \in \mathbb{N}_0$, and write $\mathcal{P} \geq \mathcal{P}'$. This defines a partial order on profiles, which forms a lattice structure.

In the following theorem we make use of the so-called “lifting” procedure, where we replace the sum of components in an extremal structure by a product of a small core and the sum of smaller structures.

Theorem 5 (Lifting Theorem). Let $\mathcal{P}$ be a biclique-profile with start index s such that $\mathcal{P}_s = t - 1$. Let $n \in \mathbb{N}$. For any cograph G on n vertices that is extremal with respect to $\mathcal{P}$ there is a finite index set I and a mapping $\sigma : I \rightarrow [t - 1]$ such that:

- (i) G admits the decomposition:

$$G = \sum_{i \in I} G_{i,1} \times G_{i,2} \quad (12)$$

where $|G_{i,1}| = \sigma(i)$ and $|G_{i,2}| \geq \sigma(i)$ for all $i \in I$.

- (ii) For any $1 \leq j \leq s - 1$, the fiber $|\sigma^{-1}(j)| \leq 1$.

We remark that in case $2 \leq s \leq t \leq 2s - 1$ (ii) and (iii) together imply that there are only finitely many connectivity components in G .

Proof of Theorem 5. Let G be a cograph on n vertices extremal with respect to $\mathcal{P}$. Since G is a cograph, each connectivity component C_i for $i \in I$ is a product. We write $C_i = G_{i,1} \times G_{i,2}$ and choose the decomposition such that $\sigma(i) := |G_{i,1}|$ is minimal. Note that $\sigma(i) < t$ must hold. If not, $|G_{i,1}| \geq t$ and $|G_{i,2}| \geq t$, implying $K_{t,t} \subseteq C_i$. As $s \leq t$, this implies $K_{s,t} \subseteq G$, which violates $\mathcal{P}_s = t - 1$.

To prove (ii), assume for a contradiction that there exist distinct indices $i_1, i_2 \in I$ such that $\sigma(i_1) = \sigma(i_2) = j$ for some $j \in [s - 1]$. Now we continue by a lifting argument.

Consider the cograph G' formed by replacing the components $C_{i_1} + C_{i_2}$ with a single component:

$$C' := E_j \times (G_{i_1,2} + G_{i_2,2} + K_j).$$

This graph G' has the same vertex count as G . We claim G' still fulfills $\mathcal{P}$.

Let $s' \geq s$ and $p := \mathcal{P}_{s'}$. It is clear that $p \geq s - 1$ as otherwise by (5) $\mathcal{P}_{s-1} < \infty$, a contradiction to the definition of s as the start index. Hence any biclique that would violate $\mathcal{P}$ has size at least $2s$.

It is clear that G' does not contain $K_{s',p+1}$ unless the product C' does contain it. However, this could only be the case if $E_j \times K_j$ does contain it, which is not possible as $2s > 2j$.

Furthermore, G' has strictly more edges than the sum $C_{i_1} + C_{i_2}$. Indeed, the cross edges between $G_{i_2,1}$ and $G_{i_2,2}$ are covered by the edges between E_j and $G_{i_2,2}$. Moreover, $\|G_{i_1,1}\| \leq K_j$ and the remaining cross edges exceed the edges inside $\|G_{i_2,1}\|$ as $\|G_{i_2,1}\| \leq \binom{j}{2} < j^2 = \|E_j, K_j\|$. This contradicts the edge-maximality of G . $\square$

3 Proofs of the Main Results

3.1 Special cases

Before we consider the general case, we prove the special cases stated in Section 1.3.

Proof of Theorem 3. Let G be a $(2, t)$ -extremal cograph on n vertices. Let $H_1, \dots, H_l$ be its connectivity components. Any of it is a product of a vertex v_j with some cograph H'_j (which might be empty in case of singletons). However, the Lifting Theorem 5 yields that there only could be one such component, yielding that G is connected and $G = v \times G'$. It is evident that the maximal degree in G' is at most $t - 1$. $\square$

Proof of Theorem 4. Let us prove the claim by induction on n . In case that $2 \leq n < 6$ it is clear that G is a clique and therefore the product of an edge with another cograph. This solves the base case.

step. We might assume that $n \geq 6$. First, let us prove that G is connected. Assume for a contradiction that $G = H_1 + H_2$ for two cographs H_1 and H_2 . Then, either $H_1 \geq 2$ and induction yields that we might replace H_1 by an edge-maximal cograph not containing $K_{3,3}$ that has a complete vertex v_1 . Otherwise, H_1 itself is a singleton and a complete vertex itself. The same argument yields a complete vertex v_2 of H_2 . However,

$$\tilde{G} := v_1 \times ((H_1 - v_1) + (H_2 - v_2) + v_2) \quad (13)$$

has one more edge than G and also does not contain $K_{3,3}$, a contradiction.

Hence, we know that $G = H_1 \times H_2$ for two cographs H_1 and H_2 on n_1 and n_2 vertices respectively. Let us assume that $n_1 \leq n_2$. Moreover, we might assume that our choices of H_1 and H_2 maximize n_1 under this restriction. It is clear that $n_1 < 3$, so to show $n_1 = 2$ we only need to show that $n_1 \neq 1$.

Let us assume for a contradiction that H_1 is made of a single vertex v . Since we assumed H_1 to be maximal under the restriction $n_1 \leq n_2$, we might assume that $H_2 = \sum_{j \in [l]} H_{2,j}$. However, it is easy to see that each $H_{2,j}$ is $(2,3)$ -extremal so Theorem 3 yields a complete vertex v_j in $H_{2,j}$. However,

$$\tilde{G} := v \times v_1 \times \left(\sum_{j \in [l]} (H_j - v_j) + \sum_{j \in [l] \setminus \{1\}} v_j \right) \quad (14)$$

does not contain $K_{3,3}$ and has $l - 1$ more edges than G , a contradiction.

Thus, we know that H_1 does contain two vertices. It is left to show that there is an extremal graph G with two complete vertices, i.e. the respective H_1 is an edge.

It is clear that H_2 does not contain a $K_{1,3}$.

Claim 1. Any edge-maximal cograph G not containing $K_{1,3}$ is the sum of cliques of size at most 3 and four-cycles.

Proof of Claim 1. It is easy to see that any connectivity component of G can have at most 4 vertices. It is easy to check manually that indeed, for any vertex count of at most 4 any edge-maximal and connected cograph not containing $K_{1,3}$ is either a clique or a four-cycle. This concludes the proof of Claim 1. $\square$

Claim 1 yields that H_2 is the sum of cliques of size at most 3 and four-cycles. We might assume that it is the sum of a triangles, b four-cycles, and a clique of size $s \in \{0, 1, 2\}$. Then,

$$\|H_2\| = 3 \cdot a + 4 \cdot b + \mathbb{1}\{s = 2\} \quad (15)$$

In case that H_2 does not contain a four-cycle, by edge-maximality it is clear that H_1 is an edge. In case that H_2 does contain a four-cycle, H_1 cannot be an edge, as otherwise a $K_{3,3}$ would arise. Let us check all cases to see that in case H_2 does contain a four-cycle, we can replace the four-cycles with cliques and H_1 by an edge and obtain a $K_{3,3}$ -free graph of the same edge count. We might replace H_2 by any edge-maximal sum of triangles, four-cycles, edges, and singletons for the given vertex count.

case $n_2 \bmod 3 = 0$. In this case, H_2 can be chosen to be made out of triangles and H_1 would be an edge.

case $n_2 \bmod 3 = 1$. In this case H_2 can be chosen to be made of triangles and either a singleton (in case $n_2 = 4$) or a single four-cycle. In the first case, H_1 is an edge. In the latter case we can replace the four-cycle by a triangle and a singleton and add the edge in H_1 for compensation.

case $n_2 \bmod 3 = 2$. In this case H_2 can be chosen to be made of triangles and either an edge (in case $n_2 = 5$) or two four-cycles. In the first case, H_1 is an edge. In the latter case, we can replace the two four-cycles by two triangles and an edge and add the edge in H_1 for compensation. This completes the proof of Theorem 4. $\square$

It is clear that $(t - 1)$ -extremal cographs are cographs of maximal degree $t - 1$. Before we study the extremal cographs, let us study regular cographs.

Lemma 2. Let $n \in \mathbb{N}$.

1. If n is even, there exists a d -regular cograph on n vertices for every $0 \leq d < n$.
2. If n is odd, there exists a $0 \leq d < n$ -regular cograph on n vertices for every even d such that $2d \neq n - 1$.

Proof of Lemma 2. We proceed by strong induction on n . The base cases $n \in \{1, 2, 3\}$ are trivial. Assume the theorem holds for all integers $k < n$. We construct a graph for n .

case 1: n is even We need to construct a cograph for any $0 \leq d < n$.

subcase 1.1: d is odd. Consider the target degree $d' = n - 1 - d$. Since n is even and d is odd, d' is even. Also $0 \leq d' < n$. By the arguments in Subcase 1.2 (below), there exists a d' -regular cograph G' . The complement $\overline{G'}$ is a cograph with degree $n - 1 - d' = d$. Thus, a d -regular cograph exists.

subcase 1.2: d is even. We work with two building blocks: K_2 and the d -regular complete multipartite cograph

$$H_d := \prod_{j \in [\frac{d}{2}+1]} E_2. \quad (16)$$

subsubcase 1.2.1: $3d \neq n - 2$. We construct $G = K_{d+1} + G'$. We need G' to be a d -regular cograph on the remaining $n' = n - (d + 1)$ vertices. Since n is even and d is even, $d + 1$ is odd, so n' is odd. Moreover,

$$3d \neq n - 2 \implies 2d \neq (n - d - 1) - 1 = 2n' - 1. \quad (17)$$

Thus, induction yields G' .

subsubcase 1.2.2: $3d = n - 2$. As the previous construction fails, we construct $G = H_d + G'$. We need G to be a d -regular cograph on $n' = n - (d + 2)$ vertices. Since n and d are even, n' is even. Either $d = 0$ and we can take $G' = E_{n'}$ or it is obvious that $2d < n - 2$ which implies that $d < n - d - 2 = n'$. In the latter case induction yields the required G' .

case 2: n is odd We need to construct a cograph for even $0 \leq d < n$ satisfying $2d \neq n - 1$.

subcase 2.1: $d < (n - 1)/2$. We construct $G = K_{d+1} + G'$. G' must be on $n' = n - (d + 1)$ vertices. Since n is odd and $d + 1$ is odd, n' is even. As $2d < n - 1$ we have $d < n - d - 1 = n'$ and induction yields G' .

subcase 2.2: $d > (n - 1)/2$. Consider the complement degree $d' = n - 1 - d$. Apparently $2d' = 2n - 2 - 2d < 2n - 2 - (n - 1) = n - 1$, so subcase 2.1 yields d' -regular cograph G' on n vertices. Consider its complement. This closes the proof of Theorem 2. $\square$

Let $t, n \in \mathbb{N}$ with $n < t$. It is clear that any $(1, t)$ -extremal graph on n vertices is a clique.

Proof of Theorem 2. In case n is even the claim is evident from Lemma 2.

We may assume that n is odd. Let G be $(1, t)$ -extremal on n vertices. Let us assume for a contradiction that there are several connectivity components $C_1, \dots, C_l$ that are not $(t - 1)$ -regular and a $(t - 1)$ -regular graph G' such that $G = \sum_{1 \leq j < l} C_j + G'$. We may assume that $l \geq 2$ is minimal among all extremal graphs.

case 1: $\sum_{1 \leq j \leq l} |C_j|$ is even. In case $\sum_{1 \leq j \leq l} |C_j| \leq t - 1$, one could replace $\sum_{1 \leq j \leq l} C_j$ by a clique, otherwise by a $(t - 1)$ -regular cograph. A contradiction to edge-maximality in both cases.

case $\sum_{1 \leq j \leq l} |C_j|$ is odd. Then $|G'|$ is even. Either $\sum_{1 \leq j < l} |C_j|$ is even or $|C_l|$ is even, we may assume the latter. Then however, also $|C_l + G'|$ is even and if $|C_l + G'| \leq t - 1$ we could replace it by a clique, otherwise by a $(t - 1)$ -regular cograph. Both are contradictions to our structural assumption.

It is clear that any connected cograph of maximal degree at most $t - 1$ has size at most $2(t - 1)$. In case it has size $2(t - 1)$ it however is $K_{t-1, t-1}$ - a $(t - 1)$ -regular cograph. This shows that $|C_1| \leq 2t - 3$. $\square$

We want to remark that for example all complete multipartite graphs with equally sized partition classes are connected regular cographs. However, there are many more, as there are even connected regular cographs with different vertex degrees in different multiplication components—consider the 4-regular cograph $2K_2 \times E_3$.

3.2 The general Pumping Theorem

To prove our Main Theorem we use the following well-known result.

Theorem 6 (Davenport). For any $n \in \mathbb{N}$, let A be a sequence of n integers. There exists a non-empty subsequence of A whose sum is divisible by n .

We now prove the Pumping Theorem stated in Section 1.3. We first introduce the cotree representation.

The *cotree* T_G of a cograph G is a rooted labeled tree representing its construction sequence. Inner vertices are labeled by $+$ or $\times$ for sum and product respectively (each of arity at least two), while leaves correspond to vertices of G . The root operation corresponds to the final step in the construction. Note that G is connected if and only if the root is labeled $\times$.

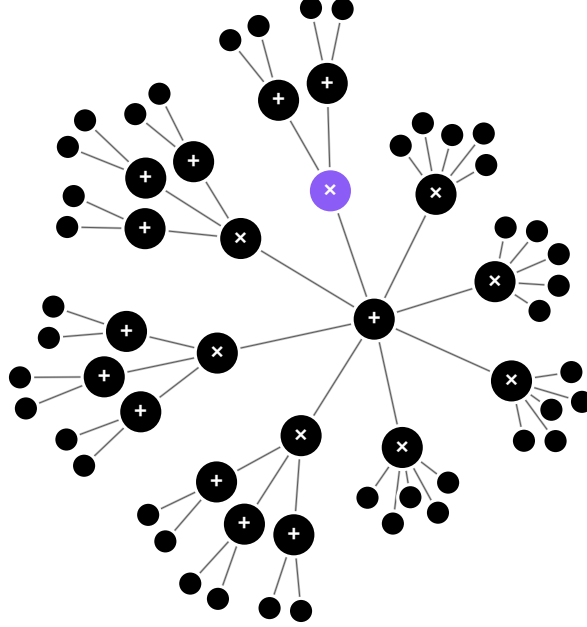


Figure 1: Cotree of a $K_{5,5}$ -free extremal cograph on 42 vertices. The root is marked as purple. The corresponding cograph is the product of $K_{2,2}$ together with the sum of four 5-cliques and three complete multipartite $K_{2,2,2}$.

For uniqueness, we require cotrees to be *reduced*: no two adjacent inner vertices share the same label (using higher-arity operations instead). The *height* of a cotree $\text{height}(T_G)$ is the maximum number of edges on any root-to-leaf path. For convenience, we write $\text{height}(G)$ directly.

The clique number behaves naturally with respect to cograph operations: $\omega(G_1 + G_2) = \max(\omega(G_1), \omega(G_2))$ and $\omega(G_1 \times G_2) = \omega(G_1) + \omega(G_2)$. Indeed, algebraically it can be seen as a valuation into the max-plus algebra. This gives the bound

$$\text{height}(G) \leq 2 \cdot \omega(G) + 1. \quad (18)$$

In particular, any (s, t) -extremal cograph has height at most $2(s + t) - 1$.

For $u \in V(T)$ let us denote the subtree of u and its descendants with respect to the root of T by $T(u)$. Let us transfer this notation to G via

$$G(u) := G[\mathcal{L}(u)], \quad (19)$$

where $\mathcal{L}(T(u))$ are the vertices that correspond to the leaves of $T(u)$. It is easy to see that $G(u)$ is the cograph corresponding to the cotree $T(u)$.

In the proof of the Pumping Theorem we are going to use the following helping Lemma.

Lemma 3 (Profile restriction). Let $\mathcal{P}$ and $\mathcal{P}^{(1)}$ be biclique-profiles. Define $\mathcal{P}^{(2)}$ by

$$\mathcal{P}_c^{(2)} := \min_{a \geq 0} \left(\mathcal{P}_{a+c} - \mathcal{P}_a^{(1)} \right) \quad (20)$$

for all $c \in \mathbb{N}_0$. Then $\mathcal{P}^{(2)}$ is a biclique-profile. Moreover, if $G = G_1 \times G_2$ and G_1 has biclique-sequence $\mathcal{S}(G_1) = \mathcal{P}^{(1)}$, then G_2 fulfills $\mathcal{P}^{(2)}$ if and only if G fulfills $\mathcal{P}$.

Proof of Lemma 3. We first verify that $\mathcal{P}^{(2)}$ is a valid biclique-profile by checking that it is monotonically decreasing and subdiagonal.

Monotonicity: Let $c < c'$. Since $\mathcal{P}$ is monotonically decreasing, $\mathcal{P}_{a+c'} \leq \mathcal{P}_{a+c}$ for all $a \geq 0$. Thus,

$$\mathcal{P}_{c'}^{(2)} = \min_{a \geq 0} \left(\mathcal{P}_{a+c'} - \mathcal{P}_a^{(1)} \right) \leq \min_{a \geq 0} \left(\mathcal{P}_{a+c} - \mathcal{P}_a^{(1)} \right) = \mathcal{P}_c^{(2)}.$$

Subdiagonality: Suppose $\mathcal{P}_j^{(2)} < l$ for some $j, l \in \mathbb{N}_0$. We need to show $\mathcal{P}_l^{(2)} < j$. By definition, there exists $a^* \geq 0$ such that $\mathcal{P}_{a^*+j} - \mathcal{P}_{a^*}^{(1)} < l$. This implies $\mathcal{P}_{a^*+j} < l + \mathcal{P}_{a^*}^{(1)}$.

By the subdiagonality of $\mathcal{P}$ applied with indices $a^* + j$ and $l + \mathcal{P}_{a^*}^{(1)}$, we obtain $\mathcal{P}_{l+\mathcal{P}_{a^*}^{(1)}} < a^* + j$.

Now, by subdiagonality of $\mathcal{P}^{(1)}$, since $a' := \mathcal{P}_{a^*}^{(1)} \geq \mathcal{P}_{a^*}^{(1)}$, we have $\mathcal{P}_{a'}^{(1)} \geq a^*$. In the minimum (20):

$$\mathcal{P}_l^{(2)} \leq \mathcal{P}_{l+a'} - \mathcal{P}_{a'}^{(1)} < (a^* + j) - a^* = j.$$

Fulfillment: Now assume $G = G_1 \times G_2$ fulfills $\mathcal{P}$ and $\mathcal{S}(G_1) \leq \mathcal{P}_1$. By Lemma 1 (ii):

$$\mathcal{S}_s(G) = \max_{a+c=s} (\mathcal{S}_a(G_1) + \mathcal{S}_c(G_2)).$$

Since G fulfills $\mathcal{P}$, for all $s \geq 0$: $\mathcal{S}_s(G) \leq \mathcal{P}_s$. Thus for any a, c with $a + c = s$:

$$\mathcal{P}_a^{(1)} + \mathcal{S}_c(G_2) \leq \mathcal{P}_s = \mathcal{P}_{a+c}.$$

Taking the minimum over all $a \geq 0$ (letting $s = a + c$ range appropriately):

$$\mathcal{S}_c(G_2) \leq \min_{a \geq 0} \left(\mathcal{P}_{a+c} - \mathcal{P}_a^{(1)} \right) = \mathcal{P}_c^{(2)}.$$

Hence G_2 fulfills $\mathcal{P}^{(2)}$. For the other direction let us assume that G_2 fulfills $\mathcal{P}^{(2)}$. By Lemma 1 (ii):

$$\mathcal{S}_c \leq \max_{a+c=s} \left(\mathcal{P}_a^{(1)} + \mathcal{P}_c^{(2)} \right) \leq \max_{a+c=s} \left(\mathcal{P}_a^{(1)} + \mathcal{P}_{a+c} - \mathcal{P}_a^{(1)} \right) = \mathcal{P}_s. \quad (21)$$

This closes the proof of Lemma 3. $\square$

Before we prove the main result, we prove a pumping Theorem for cographs extremal with respect to some biclique-profile $\mathcal{P}$. For this let us introduce the notation

$$\text{ex}(n, \{\mathcal{P}, P_4 - \text{ind}\}) \quad (22)$$

to express the edge-count of $\mathcal{P}$ -extremal cographs on n vertices. We remark that $\mathcal{P}$ -extremal cographs are exactly the edge-maximal cographs not containing a set of bicliques, specified by $\mathcal{P}$.

Theorem 7 (Pumping Theorem for biclique-profiles). Let $s, t \in \mathbb{N}$ and $\mathcal{P}$ be a biclique-profile with start index s and start value $\mathcal{P}_s = t - 1$. There exist $\alpha \in \mathbb{Q}_{\geq 0}$, $R, N_{s,t} \in \mathbb{N}$, and core graphs $G_0, \dots, G_{R-1}$ each of size at most $N_{s,t}$ with designated components $H_j \subseteq G_j$ of edge density α where any vertex in H_j has exactly the same neighbourhood outside of H_j such that for n large enough, there is a $\mathcal{P}$ -extremal cograph on n vertices obtained by pumping $H_{(n \bmod R)}$ inside of $G_{(n \bmod R)}$. That is, for $n \geq N_{s,t}$

$$\text{ex}(n, \{\mathcal{P}, P_4 - \text{ind}\}) = a_{(n \bmod R)} + \alpha n \quad (23)$$

for constants $a_0, \dots, a_{R-1} \in \mathbb{Q}_{\leq 0}$. Moreover, $\alpha \leq s - 1 + \frac{t-1}{2}$ and in case $s \geq 2$ either $\alpha < s - 1 + \frac{t-1}{2}$ or $a_0, \dots, a_{R-1} < 0$. In case that H_q is a connectivity component in G_q for some $q \in \mathbb{Z}/\mathbb{Z}$ we certainly know that $\alpha < s - 1 + \frac{t-1}{2}$.

Proof of Theorem 7. In case $s = 1$, any connectivity component of any $\mathcal{P}$ -extremal cograph G has size at most $2(t-1)$ so using Theorem 6 one can replace all but finitely many components in G by one densest component. Moreover, as the maximal degree in G is at most $t-1$ it is clear that $\alpha \leq \frac{t-1}{2}$.

Thus, let us assume that $s \geq 2$. Let us fix $D := 4t^2 + s$ and $N := D^{2(s+t)+1} + 1$. Let G be a $\mathcal{P}$ -extremal cograph on $n \geq N$ vertices. Let T_G be its reduced cotree.

1. Bounded degree of product vertices: Let u be any product vertex in T_G . Let $v_1, \dots, v_k$ be its children. Since G is a cograph, the subgraph induced by u contains a complete multipartite graph with parts of sizes $|G(v_1)|, \dots, |G(v_k)|$. Since every child in the cotree has at least one vertex, if $k \geq s + t$, one could find a $K_{s,t}$ in G . Thus, any product vertex has strictly less than $s + t \leq D$ children.

2. Existence of a high degree addition vertex: Let us assume for a contradiction that all addition vertices have at most D children. By (18) it is clear that $\text{height}(G) \leq 2(s+t) + 1$. This implies however that $n \leq D^{2(s+t)+1}$, a contradiction. Thus, we know that there exists a vertex v with at least $D + 1$ children.

3. Structure of $G(u)$ for addition vertices u : Let u be an addition vertex in T_G . All vertices in $G(u)$ have the same neighborhood W in $V(G - G(u))$. Since $|G(u)| \geq t$ it is clear that $w := |W| < s$. By Lemma 3 the subgraph $G(u)$ is extremal with respect to some biclique-profile $\mathcal{P}^{(u)}$ with start index $s - w$. By Lemma 5 (ii) the addition vertex u has at most $s - w - 1$ children $X_1^{(u)}$ such that for $x \in X_1^{(u)}$ there could be a large $G(x)$. All other children $x \in X_2^{(u)}$ have $G(x)$ of size at most $2 \cdot \mathcal{P}_{s-w}^{(u)} < 2t$.

4. Reduction to a core graph with one pumped component: Let us assume that v has $d > D$ children. Then, $|X_2^v| > d - |X_2^v| > D - s = 4t^2$. As for $x \in X_2^v$ we know $|G(x)| < 2t$ and the pigeonhole principle yields $q \in [2t]$ such that there are at least $2t$ vertices $X_{\text{pump}} \subseteq X_2^{(v)}$ such that for any $x \in X'$ we have $|G(x)| = q$. Let us denote the addition vertices with more than D children by V^+ and let us denote all the children of such addition vertices that correspond to small components except X_{pump} by

$$X_{\text{pool}} := \bigcup_{u \in V^+} X_2^{(u)} \setminus X_{\text{pump}}. \quad (24)$$

Now for any $x \in X_{\text{pool}}$ we may choose an arbitrary subset $X_{\text{test}} \subseteq X_{\text{pump}}$ of size $\mathcal{L}(x)$ and replace the vertices $\bigcup_{y \in X_{\text{test}}} \mathcal{L}(y)$ by q times pumping $G(x)$ to obtain a graph G' of the same vertex count. As all vertices in V^+ have more than t descendant leaves, we know that pumping them does not create $\mathcal{P}$ -problematic bicliques with their common neighbourhood, as otherwise there already would be a $\mathcal{P}$ -problematic biclique in G . Hence, also G' fulfills $\mathcal{P}$. This implies that

$$\|G'\| \leq \|G\|. \quad (25)$$

For $z \in V(T)$ let us introduce the notation

$$e(z) := \|G(z)\| + \|\mathcal{L}(z), V(G) \setminus \mathcal{L}(z)\|. \quad (26)$$

It is clear that for any $x_1, x_2 \in X_{\text{pump}}$ we have $e(x_1) = e(x_2)$, let us denote it by e_{pump} . Moreover, we may replace all $G(x_2)$ by one of the components, let us denote it by G_{pump} . Moreover, by (25) we have

$$q \cdot e(x) \leq |\mathcal{L}(x)| \cdot e_{\text{pump}}. \quad (27)$$

Now, for any $X_{\text{candidates}} \subseteq X_{\text{pool}}$ of size q by Davenport's Theorem 6 we find $X_{\text{replace}} \subseteq X_{\text{candidates}}$ such that q divides $\sum_{x \in X_{\text{replace}}} |G(x)|$, let us say by r . As

$$q \cdot \sum_{x \in X_{\text{replace}}} e(x) \leq e_{\text{pump}} \cdot \sum_{x \in X_{\text{replace}}} |\mathcal{L}(x)| \quad (28)$$

we conclude that

$$\sum_{x \in X_{\text{replace}}} e(x) \leq e_{\text{pump}} \cdot r. \quad (29)$$

This shows that we may replace $\bigcup_{x \in X_{\text{replace}}} \mathcal{L}(x)$ by r copies of G_{pump} without losing edges. By a similar argument as about G' we see that G_{pump} fulfills $\mathcal{P}$.

Iterating this procedure, we arrive at a $\mathcal{P}$ -extremal cograph G' , where all vertices except v have at most D children, except for at most $q - 1$ leftover children from X_{pool} . For ease of notation let us assume that $G = G'$.

Now, let us consider the graph G_{core} obtained by deleting $\bigcup_{x \in X_{\text{pump}}} \mathcal{L}(x)$. It has at most $N_{s,t} := D^{2(t+s)+1} + q \cdot 2t$ vertices. Moreover, it fulfills $\mathcal{P}$.

5. Cyclic repetition of the core graphs: For any cograph of size at most $N_{s,t}$ there is a finite number of children x of addition vertices, so that one can blow up $G(x)$ indefinitely so that the resulting cograph does not contain $K_{s,t}$. This gives a finite number of equation restriction pairs of the form

$$a_j + \alpha_j \cdot n, \quad n \bmod q_j = \gamma_j, \quad (30)$$

for $j \in \Omega$ that correspond to a cograph G_j on at most $N_{s,t}$ vertices together with the child of an addition vertex $u_j \in V(G_j)$ that's corresponding subgraph $G(u)$ would be pumped. Observe that $q_j = |\mathcal{L}(u_j)|$ and $\gamma_j = |G_j| \bmod q_j$. We conclude that for $n \geq N_{s,t}$

$$\text{ex}(n, \{K_{s,t}, P_4 - \text{ind}\}) = \max \{a_j + \alpha_j \cdot n \mid j \in \Omega, \quad n \bmod q_j = \gamma_j\}. \quad (31)$$

As for large n the linear part $\alpha_j \cdot n$ takes over the constant a_j there is N, R and $\sigma : \mathbb{Z}/R\mathbb{Z} \rightarrow \Omega$ such that for any $n \geq N$

$$\text{ex}(n, \{K_{s,t}, P_4 - \text{ind}\}) = a_{\sigma(n \bmod R)} + \alpha_{\sigma(n \bmod R)} \cdot n. \quad (32)$$

Moreover, we know that there is α such that for any $j \in \mathbb{Z}/R\mathbb{Z}$ we have $\alpha_j = \alpha$. Otherwise, there are distinct $j_1, j_2 \in \mathbb{Z}/R\mathbb{Z}$ such that $\alpha_{j_1} < \alpha_{j_2}$. We could "shift" the pumped up $G_{\sigma(j_2)}$ into the vertex count modulo class j_1 with respect to R by adding singletons. For large n this leads to a contradiction, as the linear term of the extremal number surpasses the constant term.

6. Upper bound on the extremal function. Let us assume for a contradiction that there is $q \in \mathbb{Z}/R\mathbb{Z}$ such that $a_q > 0$. Consider a $\mathcal{P}$ -extremal graph G on n vertices with $n \bmod R = q$ that was obtained by pumping up H_q in G_q . Now, $(R+1) \cdot G$ still fullfills $\mathcal{P}$ but has more edges than the extremal graph obtained by pumping H_q to the same vertex count, a contradiction.

Let us proceed proving the claimed upper bound of α and $a_0, \dots, a_R$ by induction on the lexicographic order of $(s, t-1)$.

In case $s = 1$ the bound $\alpha \leq \frac{t-1}{2}$ follows from the maximum degree condition on the $\mathcal{P}$ -extremal cographs.

In case $s = 2$ it is clear that $\text{ex}(n, \{\mathcal{P}, P_4 - \text{ind}\}) \leq \text{ex}(n, \{K_{2,t}, P_4 - \text{ind}\})$. However, we know the shape of $(2, t)$ -extremal cographs by Theorem 3: They are the product of a vertex with a $(1, t)$ -extremal cograph, giving the bound

$$\text{ex}(n, \{K_{2,t}, P_4 - \text{ind}\}) \leq (n-1) \cdot \left(s-1 + \frac{t-1}{2}\right), \quad (33)$$

proving that either $\alpha < s-1 + \frac{t-1}{2}$ or $a_0, \dots, a_R < 0$.

Let us assume $s \geq 3$ and the statement holds for any biclique-profile with lexicographically smaller $(s, t-1)$. Let $q \in \mathbb{Z}/R\mathbb{Z}$.

case 1. H_q is not a connectivity component in G_q . Let W be the common neighborhood of H_q in G_q and $w := |W| \geq 1$.

By Lemma 3 we know that there is a biclique-profile $\mathcal{P}'$ with start index $s-w$ and start value $\mathcal{P}'_{s-w} = t-1$ such that H_q is $\mathcal{P}'$ -extremal. If H_q would have (internal) edge-density greater than $s-w-1 + \frac{t-1}{2}$, we could consider $m \cdot H_q$ demonstrating that for $\mathcal{P}'$ the linear term of $\text{ex}(n, \{\mathcal{P}', P_4 - \text{ind}\})$ is greater than $s-w-1 + \frac{t-1}{2}$, a contradiction to the induction assumption.

Hence we know that the edge-density of G_q is at most $s-w-1 + \frac{t-1}{2}$. As any vertex in H_q has exactly w neighbors outside of H_q this immediately implies that $\alpha \leq s-1 + \frac{t-1}{2}$.

Let again G be an extremal graph obtained by pumping H_q in G_q . Let us assume for a contradiction that $a_q = 0$, i.e. the edge density of G is α . Let M be the connectivity component of G that contains the pumped up H_q . It is clear that M is the product of a cograph on at most $s-1$ vertices with another cograph. Now, $(R+1) \cdot G$ is a $\mathcal{P}$ -extremal cograph by edge count that contains many copies of M , a contradiction to the Lifting Theorem 5.

Hence, we now that the edge density of G_q is less than α and therefore $a_q < 0$.

case 2. H_q is a connectivity component of G_q , i.e. $H_q = M_1 \times M_2$ for two cographs M_1, M_2 on m_1, m_2 vertices respectively.

As there are $\mathcal{P}$ -extremal cographs where H_q is pumped up by the Lifting Theorem 5 we know that $s \leq m_1, m_2 \leq t-1$. By Lemma 3 know that there is a biclique-profile $\mathcal{P}^{M_1}$ such that M_1 is $\mathcal{P}^{M_1}$ -extremal. We also know that M_1 does not contain $K_{s,t-m_2}$.

case $s \leq t - m_2$. To show an upper edge bound we may weaken the biclique-profile $\mathcal{P}^{M_1}$ by setting values at indices less than s to infinity. In case that the edge density of M_1 is at least $s - 1 + \frac{t - m_2 - 1}{2}$, we might clone M_1 , resulting in a bound $\text{ex}(\mathcal{P}^{M_1}, P_4 - \text{ind}) \geq (s - 1 + \frac{t - m_2 - 1}{2}) \cdot n$ for infinitely many n , contradicting the induction assumption for $(s, t - m_2 - 1)$ as $s \geq 2$. We deduce

$$\text{avgdeg}_{M_1}(M_1) < 2(s - 1) + t - m_2 - 1 \quad (34)$$

Hence, the average degree of vertices $V(M_1)$ inside H_q is upper bounded by

$$\text{avgdeg}_{H_q}(M_1) < 2(s - 1) + t - 1. \quad (35)$$

case $t - m_2 \leq s$. To show an upper edge bound we may weaken the biclique-profile $\mathcal{P}'$ by setting values at indices less than $t - m_2$ to infinity. A similar argument using the induction hypothesis for $(t - m_2, s - 1)$ yields

$$\text{avgdeg}_{M_1}(M_1) \leq 2(t - m_2 - 1) + s - 1 \quad (36)$$

Hence, the average degree of vertices $V(M_1)$ inside H_q is upper bounded by

$$\text{avgdeg}_{H_q}(M_1) \leq 2(t - m_2 - 1) + s - 1 + m_2 \leq 2(s - 1) + t - 1, \quad (37)$$

where we used our assumption $t - m_2 \leq s$. Similarly to before, in case $t - m_2 \geq 2$ we also obtain the inequality

$$\text{avgdeg}_{H_q}(M_1) < 2(s - 1) + t - 1. \quad (38)$$

By symmetry we also see that $\text{avgdeg}_{H_q}(M_2) \leq 2(s - 1) + t - 1$ with an inequality in case $t - m_1 \geq 2$. In case that either $m_1 < t - 1$ or $m_2 < t - 1$ this shows $\alpha < s - 1 + \frac{t - 1}{2}$.

Thus, we may assume that both $m_1 = m_2 = t - 1$. However, in this case the maximal degree within M_1 and M_2 is at most $s - 1$, resulting in a total edge density of H_q of at most $\frac{s - 1 + t - 1}{2}$. Again we showed that $\alpha < s - 1 + \frac{t - 1}{2}$. To summarize, either there is a q such that H_q is a connectivity component. In this case, $\alpha < s - 1 + \frac{t - 1}{2}$. Otherwise, for all q the H_q have common neighbours and we proved that $a_q < 0$. This closes the proof of Theorem 7. $\square$

Proof of Theorem 1. Let us define $\mathcal{P}$ by

$$\mathcal{P}_j := \begin{cases} \infty & j < s \\ t - 1 & s \leq j < t \\ s - 1 & j \geq t \end{cases} \quad (39)$$

Theorem 7 yields $R, N_{s,t} \in \mathbb{N}$ and core graphs $G_0, \dots, G_{R-1}$ each of size at most $N_{s,t}$ with designated components $H_j \subseteq G_j$ of edge density α such that for any n large enough, there is a $\mathcal{P}$ -extremal cograph on n vertices obtained by pumping $H_{(n \bmod R)}$ inside of $G_{(n \bmod R)}$. Consider

$$\tilde{G}(r) := K_{s-1} \times (r \cdot K_t) \quad (40)$$

It is clear that $\tilde{G}(r)$ does not contain $K_{s,t}$ and

$$n_r := |\tilde{G}(r)| = s - 1 + r \cdot t \quad (41)$$

$$\|G(r)\| = \binom{s-1}{2} + \left(s - 1 + \frac{t-1}{2}\right) \cdot (n_r - s + 1) \quad (42)$$

This proves that $\alpha \geq s - 1 + \frac{t-1}{2}$ and by the bound given by Theorem 7 we conclude

$$\alpha = s - 1 + \frac{t-1}{2}. \quad (43)$$

We know from the proof of Theorem 7 that the pumping components H_q can not be connectivity components in G_q as otherwise we would have an inequality here. Hence, H_q has a common neighborhood W in G_q of size $w := |W| \geq 1$.

It is clear that $w \leq s$. Let us assume for a contradiction that $w \leq s - 2$. Then by Lemma 3 there is a biclique-profile $\mathcal{P}'$ with start-index $s - w \geq 2$ and start value $t - 1$ such that $m \cdot H_q$ is $\mathcal{P}'$ -extremal for any $m \in \mathbb{N}$. However,

again Theorem 1 yields for the linear coefficient α' of $\text{ex}(n, \{\mathcal{P}', P_4 - \text{ind}\})$ since H_q is a connectivity component in $m \cdot H_q$ that $\alpha' < s - w - 1 + \frac{t-1}{2}$, a contradiction to $\alpha = s - 1 + \frac{t-1}{2}$.

This proves that any vertex in H_q has $s - 1$ vertices outside of H_q and the edge density of H_q is $\frac{t-1}{2}$ accordingly. By the equality $\alpha = s - 1 + \frac{t-1}{2}$ again Theorem 7 yields that the constants $a_0, \dots, a_R < 0$.

Let $q \in \mathbb{Z}/R\mathbb{Z}$. Then for large n with $n \bmod R = q$ one can obtain an (s, t) -extremal cograph G on n vertices by pumping up H_q in G_q . Let G'_q be the connectivity component of G_q that contains H_q . Then in G the connectivity component obtained from pumping H_q in G'_q is itself (s, t) -extremal. Hence for any n large enough such that $n + |G_q - G'_q| \bmod R = q$ there is an (s, t) -extremal connected cograph on n vertices and one could pick at least one residue class such that the respective extremal cographs are connected. This closes the proof of Theorem 1. $\square$

4 Conclusion

We have shown that the bipartite Turán problem on cographs admits an exact linear solution for large n , characterized by a periodic "pumping" of specific dense components. This contrasts with the general bipartite Turán problem, where bounds are polynomial but not linear. The strong structural constraints of cographs (bounded clique-width, bounded VC-dimension) force the extremal graphs into a rigid algebraic shape. While it was known that the $\text{ex}(n, \{K_{s,t}, P_4 - \text{ind}\})$ can be linearly bounded in n by a result of Bonamy et al. [4], we are the first recognizing the cyclic behaviour and the determining the linear coefficient of extremal function for large n to be $s - 1 + \frac{t-1}{2}$.

Finally, we want to remark that the exact determination of $\text{ex}(n, \{H, P_4 - \text{ind}\})$ is still an unsolved and interesting problem for H that is neither a clique nor a biclique.

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